



Figure 3: Arduino Mega

- 1) Install all pre-requisites as shown below:

```
sudo apt-get install build-essential
sudo apt-get install cmake
sudo apt-get install pkg-config
sudo apt-get install libpng12-0 libpng12-dev libpng++-dev libpng3
sudo apt-get install libpnglite-dev libpngwriter0-dev
libpngwriter0c2
sudo apt-get install zlib1g-dbg zlib1g zlib1g-dev
sudo apt-get install libjasper-dev libjasper-runtime libjasper1
sudo apt-get install pngtools libtiff4-dev libtiff4 libtiffxx0c2
libtiff-tools
sudo apt-get install libjpeg8 libjpeg8-dev libjpeg8-dbg libjpeg-
progr
sudo apt-get install ffmpeg libavcodec-dev libavcodec52
libavformat52 libavformat-dev
sudo apt-get install libgstreamer0.10-0-dbg libgstreamer0.10-0
libgstreamer0.10-dev
sudo apt-get install libxine1-ffmpeg libxine-dev libxine1-bin
sudo apt-get install libunicap2 libunicap2-dev
sudo apt-get install libdc1394-22-dev libdc1394-22 libdc1394-utils
sudo apt-get install swig
sudo apt-get install libv4l-0 libv4l-dev
sudo apt-get install python-numpy
sudo apt-get install build-essential libgtk2.0-dev libjpeg62-
dev libtiff4-dev libjasper-dev libopenexr-dev cmake python-
dev python-numpy libtbb-dev libeigen2-dev yasm libfaac-dev
libopencore-amrnb-dev libopencore-amrwb-dev libtheora-dev
libvorbis-dev libxvidcore-dev
```

- 2) Install Python development headers with `sudo apt-get install python-dev`.
- 3) Download the source code from <http://sourceforge.net/projects/opencvlibrary/files/opencv-unix/2.3/>. It is recommended that you move the downloaded OpenCV package to your `/home/<user>` directory. With that as your current directory, extract the archive, as follows:

```
tar -xvf OpenCV-2.3.1a.tar.bz2
cd OpenCV-2.3.1/
```

- 4) Make a new directory called `build` and `cd` into it (`mkdir build; cd build`).
- 5) Run Cmake:

```
cmake -D WITH_TBB=ON -D BUILD_NEW_PYTHON_SUPPORT=ON -D WITH_
V4L=OFF-D INSTALL_C_EXAMPLES=ON -D INSTALL_PYTHON_EXAMPLES=ON -D
BUILD_EXAMPLES=ON ..
```

- 6) Follow this with `make` and `sudo make install` (to install the library).
- 7) Next, configure the system to use the new OpenCV shared libraries. Edit a configuration file with `sudo gedit /etc/ld.so.conf.d/opencv.conf`. Add the following line at the end of the file (it may be an empty file, which is okay) and then save it:

```
/usr/local/lib
```

Close the file and run `sudo ldconfig`.

- 8) Open your system `bashrc` file (for me, that is `sudo gedit /etc/bash.bashrc`) and add the following code at the end of the file:

```
PKG_CONFIG_PATH=$PKG_CONFIG_PATH:/usr/local/lib/pkgconfig
export PKG_CONFIG_PATH
```

- 9) Save and close the file, then log out and log in again, or reboot the system.

I have made a video tutorial for the installation of OpenCV 2.3 with Python support, which you can view at <http://www.youtube.com/watch?v=b3IQyhJXK78>

In the second part of this series, I plan to cover the basics and functions of OpenCV, and then how to develop image-processing programs based on it. In the third (and final) part, I will cover Arduino and Pyserial programming, as well as how to integrate all five components to finally build an embedded system for image processing. 

Further reading

- [1] <http://http://opencv.itseez.com/>
- [2] <http://pyserial.sourceforge.net/>
- [3] <http://www.arduino.cc/>
- [4] <http://pandaboard.org/>

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The author is a FOSS advocate and loves to explore different open source technologies. His areas of interest include OpenCV, Python, Android, Linux, Arduino and other open source hardware platforms. In his spare time, he likes to make video tutorials on various open source technologies, which can be found on YouTube. He is also a big Arsenal fan and can be reached at jayneil.dalal@gmail.com